Simulation Arena

Discount Leakage & Cherry-Picking: The Basket Sabotage

Cherry-pick risk 50/100 (Low): give this discount and there is a 44.0% chance the Support line or another offsetting line disappears, costing $50k - 10.2% of the deal's profit.

Seed 42 | Run run_6b346374ceec | 2,000 scenarios tested | Cluster bootstrap resampling with empir… | 22 Sep 2026, 08:24

Same seed, same numbers - this run can be reproduced exactly.

How this works

We went back through your past multi-line deals and found the ones where a line was discounted past 30%. In those deals we measured how often the other lines were cancelled inside 6 months, and compared it with how often those lines die anyway. That difference is the cherry-pick signal. We then rebuilt this quote 2,000 times, each time re-drawing whole past deals so the answer reflects how much history we actually have, and re-priced the basket with whatever survived. The risk score is three things you can check by hand: how far past your discount limit the ask sits (25%), how likely the offsetting lines are to vanish (50%), and how much of the profit sits in the line being cut (25%).

The numbers that matter

CHERRY-PICK RISK SCORE **50 / 100** Built from three checkable pieces; the biggest one is how likely the offsetting lines are to vanish. vs Low risk	PROFIT LIKELY TO LEAK AWAY **$50k** The profit this deal loses once you allow for how buyers actually behave. vs 10.2% of the deal's profit	CHANCE THE SUPPORT LINE IS DROPPED **13.6%** How often the line that pays for the discount dies in the first 6 months. vs 1.2x the normal 11.8%
CHANCE ANY OFFSETTING LINE DIES **44.0%** The odds the basket does not survive intact.	PROFIT EXPOSED ON THE SUPPORT LINE **$20k** $150k of profit with a 13.6% chance of walking.	PROFIT AT RISK **10.2%** Share of the bundle's profit that is exposed to cherry-picking.
PROFIT IF THEY KEEP EVERYTHING **$489k** The spreadsheet number: 54.3% margin on $900k.	PROFIT YOU SHOULD ACTUALLY PLAN FOR **$439k** The honest number after cherry-picking. vs as low as $300k in a bad run	HOW MUCH MORE OFTEN THE WORST PAIR DIES **1.9x** Deals with a deep Support discount lose the Software line 1.9 times more often than usual. vs 9.1% vs 4.8%
CHANCE THE BUNDLE ENDS UNDER YOUR 35% FLOOR **0.8%** How often the surviving basket is not worth signing at all.		

How we got there

1. What is actually on the table

The basket is worth $900k and makes $489k if nothing is dropped. But 61.3% of that profit sits in Software - the exact line the buyer wants 35% off, which is 5 points past the discount line you set yourself.

Evidence: $300k of $489k profit in Software

2. What history says happens next

In 22 past deals where Support was discounted past 30%, the Software line was cancelled 9.1% of the time against a normal rate of 4.8% - 1.9 times more often. And they do not do it immediately - when a line dies it happens around month 3.

Evidence: source: your own history

3. What the rebuild shows

Rebuilding this quote 2,000 times, the profit lands at $439k on average, and in an unlucky run of twenty it is $300k or worse. There is a 0.8% chance the bundle that survives is worth less than your 35% margin floor.

Evidence: $50k average shortfall

4. What to do about it

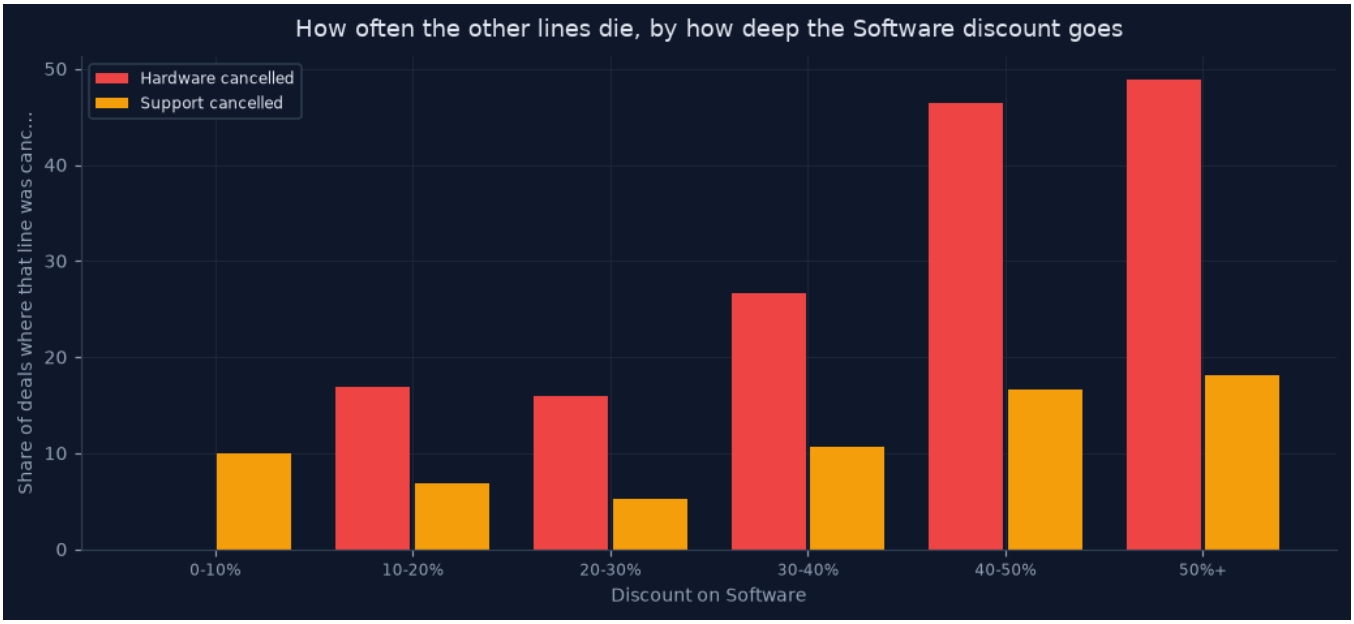
The biggest driver of the score is how likely the offsetting lines are to vanish. Tie the discount to the behaviour: if the Support line is not live at 80% of committed hours by month 6, the 35% Software discount comes off and Software reverts to standard pricing - worth up to $210k back.

Evidence: clawback up to $210k

Charts

How often the other lines die, by how deep the Software discount goes

Each bar is a group of past deals with a similar discount



What this shows: If the bars jump once the discount passes your limit, buyers are not just buying more - they are buying differently. That jump is the cherry-pick signal.

Which discount poisons which line

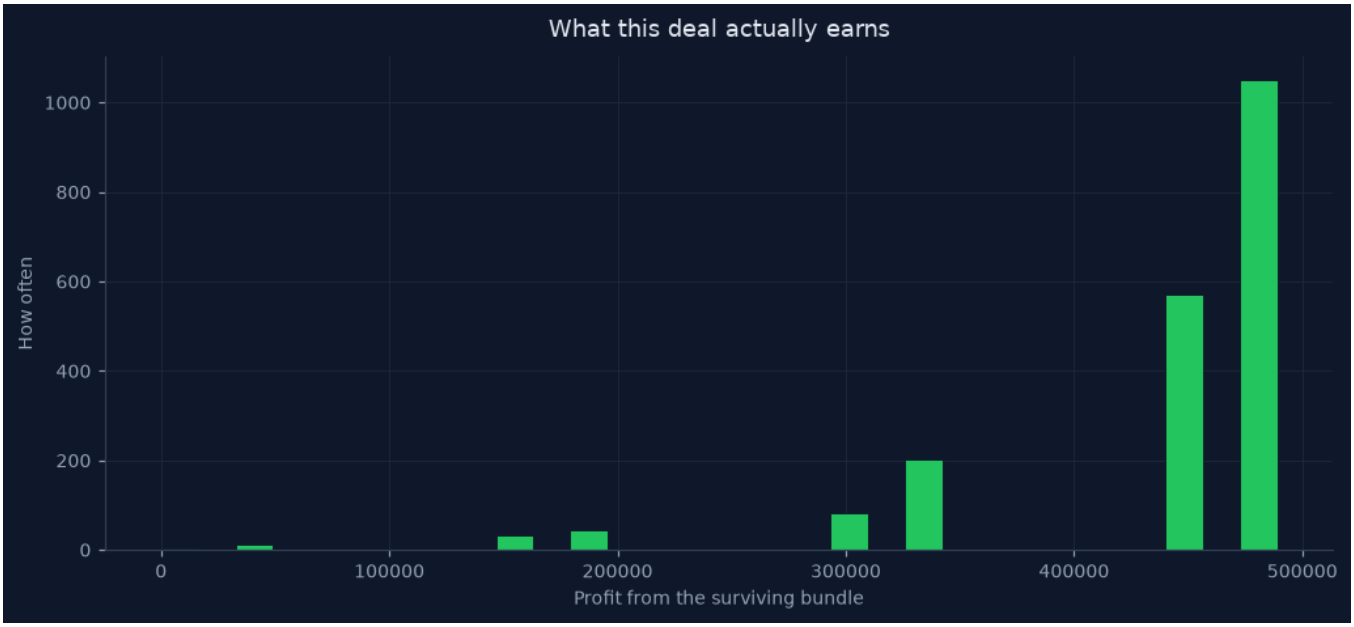
Numbers are 'times more likely than usual' when the row's line is discounted past 30% - rows with too little history are left out



What this shows: Anything above 1.0 means the two events travel together. The hottest cell is the pair you should write a contract clause about.

What this deal actually earns

2,000 rebuilt bundles - the dashed line is the number on the quote



What this shows: The gap between the tall bars and the 'if nothing is dropped' marker is the cost of cherry-picking. The left tail is the bad case you are signing up for.

Tables

The quote, line by line

Where the profit sits, and how exposed each line is.

Line	List price	Discount	Price after discount	Your margin	Profit	Share of profit	Chance it is dropped	Profit at risk
Software	$600,000	35.0%	$390,000	85.0%	$300,000	61.3%	4.8%	$14,341
Hardware	$300,000	5.0%	$285,000	18.0%	$39,000	8.0%	36.0%	$14,055
Support	$250,000	10.0%	$225,000	70.0%	$150,000	30.7%	13.6%	$20,352

Row 1 carries the discount being asked for.

What happens to each line when another line is discounted past 30%

Read across a row: 'when THIS line is deeply discounted, the chance THAT line dies'. A figure above 1.0 means the two events travel together.

Discounted line (A)	Line that dies (B)	Deals like this	Normal rate	Rate after the discount	Times more often	Link strength
Hardware	Hardware	789	31.6%	32.2%	1.02x	none
Hardware	Services	789	58.4%	57.5%	0.99x	none
Hardware	Software	789	4.8%	3.7%	0.77x	none
Hardware	Support	789	11.8%	9.4%	0.80x	none
Services	Hardware	144	31.6%	31.2%	0.99x	none
Services	Services	144	58.4%	55.6%	0.95x	none
Services	Software	144	4.8%	5.6%	1.17x	none
Services	Support	144	11.8%	11.8%	1.00x	none
Software	Hardware	7,761	31.6%	36.0%	1.14x	clear
Software	Services	7,761	58.4%	65.8%	1.13x	clear
Software	Software	7,761	4.8%	4.8%	1.01x	none
Software	Support	7,761	11.8%	13.6%	1.15x	weak
Support	Hardware	22	31.6%	31.8%	1.01x	none
Support	Services	22	58.4%	45.5%	0.78x	none
Support	Software	22	4.8%	9.1%	1.91x	none
Support	Support	22	11.8%	54.5%	4.64x	weak

Rows where the second line dies far more often than normal are the cherry-pick signature.

What the bundle actually earns

Every one of the 2,000 rebuilt bundles, summarised.

Scenario	Profit	Margin	Versus the plan
If they keep everything	$489,000	54.3%	the number on the quote
Most likely outcome	$489,000	54.3%	$0
A bad run (1 in 20)	$300,000	33.3%	-$189,000
Worst bundle we saw	$0	0.0%	-$489,000

Row 2 is the number to plan with.

Does the buyer's tier change the behaviour?

Same test, split by customer tier (only tiers with enough history are shown).

Customer tier	Deals	Deals with a deep discount	Chance each offsetting line dies
Enterprise	2,805	2,205	36%, 15%
Global-2000	1,135	857	36%, 13%
Mid-Market	3,596	2,774	35%, 13%
SMB	2,464	1,925	37%, 13%

Use the worst tier as your planning number if you do not know the buyer well.

What we recommend

Approve the 35% only as a conditional discount - expect to lose $50k if you do not tie it to the Support line.

Risk Low 50 out of 100 | Confidence: High

Why

- 61.3% of this deal's profit sits in Software, the line being cut by 35%.
- Past deals like this one lose the offsetting lines 44.0% of the time (your own history).
- Expected profit after cherry-picking is $439k, not $489k - a $50k gap.

What to do

1. Make the 35% Software discount conditional: if the Support line is not live at 80% of committed hours by month 6, Software pricing reverts to the standard tier (recovering up to $210k).

2. Do not let them drop one offsetting line and keep the other: make the discount conditional on the whole Hardware, Support basket surviving, not on any single line.

3. Put the test at month 6, not at signature - past buyers who walk do it around month 3.

The trade-off: Walking away from the 35% risks the deal; giving it unprotected costs about $50k. The clause lets you keep the deal and cap the loss. If the buyer refuses any condition, the mildest version is to cap Software at 30%, which still protects $30k.

What we assumed and what to watch

Worth knowing before you act

• Some category pairs rest on fewer than 30 past deals, so those figures are rough. The headline pair is the one with enough history behind it.

• Your file holds 2,251,151 rows. We are running on an evenly spread sample of 225,024 of them (10.0%), keeping whole records (grouped by your 'quote_id' column) intact rather than splitting a record down the middle, because a working set that big does not fit in memory while keeping the tool responsive. Every number below comes from those 225,024 rows.

Assumptions

• Quote under review: Software $600,000 at 35% off (85.0% margin), Hardware $300,000 at 5% off (18.0% margin), Support $250,000 at 10% off (70.0% margin).

• Deep discount means anything past 30%; a line counts as abandoned if it dies within 6 months.

• Abandonment rates come from your own history: 10,000 past deals, 8,716 deep-discount category observations.

• Risk score = 25% x how far past your discount limit the ask sits (10 points past = full marks) + 50% x chance an offsetting line dies + 25% x share of profit sitting in the discounted line.

• A cancelled line is treated as fully lost: no revenue and no cost.

• Bundle margin floor 35%; minimum support commitment 80% of hours.

• We replayed 2,000 bundles built from your own past deals; seed 42 - re-running gives identical numbers.

• Quote / deal ID -> your 'quote_id' column

• Customer tier / segment -> your 'customer_tier' column

• Line item ID -> your 'line_id' column

• Product line name -> your 'product_line' column

• Line category -> your 'category' column

• List price of the line -> your 'list_price' column

• Cost of the line -> your 'cost' column

• Discount on the line (%) -> your 'discount_pct' column

• Was the line cancelled? (1/0) -> your 'cancelled_within_6m' column

• Months until cancellation -> your 'months_to_cancel' column

• Data used: 225,024 of the 2,251,151 rows in your file (10.0%), drawn evenly across it.

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
1.33s	prepare	Cost of the line -> your 'cost' column
1.33s	prepare	Discount on the line (%) -> your 'discount_pct' column
1.33s	prepare	Was the line cancelled? (1/0) -> your 'cancelled_within_6m' column
1.33s	prepare	Months until cancellation -> your 'months_to_cancel' column
6.68s	prepare	Read 225,024 records into the model.
6.68s	simulate	Running the simulation…
6.68s	read	Reading the quote you want us to stress-test...
6.68s	read	The basket is worth $900k and carries $489k of profit (54.3% margin) if the buyer keeps every line.
6.68s	read	The line being discounted hardest is Software at 35% off - that is 5 points past your 30% deep-discount line.
7.11s	read	Learning from 10,000 past deals across 4 categories (Hardware, Services, Software, Support)...
7.11s	read	Your file holds 49,999 past deals. Rebuilding every one of them thousands of times needs more memory than a browser-class machine has, so the rebuilding works on an evenly spread pool of 10,000 of them (20.0%). Every nu…
7.18s	prepare	Checking what happened in past deals that got a discount this deep...
7.18s	prepare	Found it: when Support is discounted past 30%, the Software line is cancelled 9.1% of the time - 1.9x its normal rate.
7.18s	prepare	Chance the offsetting lines disappear: 44.0% - that is what the discount is really buying.
7.18s	simulate	Rebuilding 2,000 bundles that look like this one and re-pricing each of them...
7.76s	simulate	Rebuilt 10% of the bundles...
7.76s	simulate	Rebuilt 20% of the bundles...
7.76s	simulate	Rebuilt 30% of the bundles...
7.76s	simulate	Rebuilt 40% of the bundles...
7.76s	simulate	Rebuilt 60% of the bundles...
7.76s	simulate	Rebuilt 70% of the bundles...
7.76s	simulate	Rebuilt 80% of the bundles...
7.76s	simulate	Rebuilt 90% of the bundles...
7.76s	summarize	All bundles priced. Working out what this deal really earns...
7.77s	summarize	Done. Cherry-pick risk 50/100 (Low) - expect to lose $50k unless the discount is made conditional.